High-Throughput FPGA-Compatible TRNG Architecture Exploiting Multistimuli Metastable Cells

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***Abstract*— This paper presents a True Random Number Generator (TRNG) exploiting latched-XOR (LX) gates and its implementation on a Xilinx Spartan 6 FPGA device. The proposed LX-TRNG aims at improving the Throughput (TP) of conventional ring oscillators (ROs) based TRNGs by combining the effect of latches metastability and ROs jitter. Measurements results have demonstrated that the generated bitstreams show very good randomness exhibiting a byte (bit) entropy of 7.9979 (0.9997), according to T8-test of AIS-31. The proposed TRNG has also been extensively tested under voltage and temperature variations showing very good robustness. In particular both NIST's and AIS-31 tests are passed for all the considered supply voltage and temperature ranges. The FPGA implementation occupies only 9 Slices and, despite its compactness, it exhibits a throughput as high as 12.5 Mbit/s with a 50 MHz operating frequency. The computation of the figure of merit FOM_E has shown the capability of the proposed TRNG to optimize the trade-off between hardware resources, bitstreams entropy and throughput, outperforming previous works.**

***Index Terms*— TRNGs, field programmable gate array (FPGA), ring oscillators, metastability, jitter.**

I. Introduction

TRUE Random Number Generators are among the most challenging and useful building blocks in cryptographic applications and cover a wide range of application fields. In detail, TRNGs are cryptographic primitives capable to generate non-deterministic bitstreams by leveraging on entropic sources, and represent a critical building block in many applications. Indeed, many modern authentication protocols, such as privacy preserving mutual authentication (PPMA), employ TRNGs as nonce sources [1], [2], [3]. TRNGs play also a key role to generate keys and tokens [4], as well as random masks used against side channel attacks [5]. Moreover, TRNGs are widely exploited in software problems like Monte-Carlo simulations [6], [7].

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On the other hand, one of the main drawbacks of these cryptographic primitives is the hardware resource consumption that is even more demanding due to the more and more restrictive requirements that should be addressed by TRNGs both in terms of randomness and speed.

FPGAs are among the most used platforms to implement this kind of cryptographic functions and stand out for their cost-effectiveness and variety of available building blocks. The FPGA implementation of high performance TRNGs able to generate bitstreams with good reliability can be a very challenging task and in many cases heavy post-processing techniques (such as Von Neumann encoding and hashing functions) have to be adopted in order to pass randomness tests (e.g. AIS-31 and NIST test suites).

The physical entropy sources usually exploited in TRNGs primitives are the jitter accumulation [8], [9], [10], the metastability [11], [12], [13] and, finally, thermal, shot and flicker noise [14]. These sources of entropy can be exploited in several ways and, as a consequence, several TRNG architectures have been proposed in the literature. For instance, the Coherent Sampling (CS) technique can be exploited to extract the jitter from two ROs which oscillate at frequencies that are mutually prime [15]. Many TRNG architectures in the literature are based on Ring-Oscillator (RO) configurations [5], [9], [16] and many techniques to improve the throughput have been proposed, such as, for example, the programmable delay line approach in [8]. TRNGs based on metastable cells are a valuable alternative, and can reach even higher throughput than RO-based ones [11], [12]. In addition, the fast carry logic of Xilinx's FPGAs can be exploited to improve the ROs' frequencies and improve the TP of Jitter-based TRNGs. However, such designs may turn out to be not trivial and tedious routing strategies should be taken into account [17]. The Multi-Stage-Feedback Ring Oscillator (MSFRO) TRNG aims at overcoming these limitations by exploiting feedback strategies to generate high frequency ROs affected by Jitter which samples an high frequency signal given by a PLL, thus generating very high TP TRNGs. However, it is worth noting that, high throughput TRNGs can result in higher resource usage and may exhibit low entropy and, as a consequence, a trade-off between hardware resource usage, bitstreams quality and throughput has to be addressed. Moreover, the bitstreams quality of high TP architectures can be sensitive to environmental variations and, in many cases particular care has to be devoted to guarantee sufficient robustness of the TRNG primitives. To reduce the hardware footprint, the Digital Clock Manager (DCM) of Xilinx architectures can be employed in Beat Frequency Detection (BFD) operation [18], [19]. An approach based on Latched Ring Oscillators (LROs) has been recently proposed to drastically reduce the hardware consumption [10]. However, lightweight architectures generate bitstreams which may not satisfy randomness requirements and, as a consequence, design strategies aiming at improving bitstreams' behaviors have to be taken into account. For instance, a simple and easy portable TRNG was presented in [20]: despite its FPGA-friendliness it requires post-processing techniques to improve the randomness. In [21] a design that aims to overcome post-processing strategies was presented, but, due to its low entropy, it has revealed vulnerabilities to frequency injection (F.I.) which was exploited in [22] to carry out a successful attack. However, the ease of implementation of RO-based TRNG has motived researchers to find new strategies to improve randomness performances without post-processing techniques. For example, the oscillation frequency of ROs can be modulated by exploiting delay-lines control [8], the excitation time of metastable architectures can be changed by adopting feedback strategies [10] or the effect on randomness of different frequencies in BFD operation can be exploited in DCM-based TRNGs [18], [19]. In particular, DCM-based TRNGs are fascinating due to their simple architecture and easy implementation but they may lack in good entropy generation. Also the MSFRO TRNG may present limitations in terms of entropy, and parity filters should be considered in order to improve the bias of the bitstream.

In this work we propose an innovative TRNG architecture which leverages on the noise characteristics of latched XOR-gates to exploit both jitter and metastability. By combining multiple entropy sources, the proposed approach results in a compact and very high throughput TRNG. The TRNG has been implemented on the Xilinx Spartan-6 XC6SLX16 FPGA and to get the most of the FPGA architecture a thorough study of building blocks and connections has been conducted. The computation of the figures of merit FOM (introduced in [10]) and FOM_E (introduced in this work) has pointed out that the proposed TRNG architecture allows to optimize the trade-off between hardware resources, bitstreams quality (in terms of entropy evaluated through NIST and AIS tests) and throughput, outperforming previous works. With respect to our previous work [10], this paper exploits a different basic cell for entropy generation (made up of only logic gates), and presents a novel approach to combine a number of low entropy basic cells to develop TRNGs in which the tradeoff between throughput, overall entropy and resources usage can be adjusted. Moreover, in this work we have explained in detail the design flow and criteria which have been taken into account to reach the optimal tradeoff, providing also a mathematical justification of the XOR combining approach which can be used in general to reach optimal design in terms of entropy and TP starting from different types of low entropy basic cells.

The paper is organized as follows. The novel TRNG architecture and its principle of operation are introduced in Section II, whereas the design and FPGA implementation are detailed in Section III. Measurement results of NIST's test suite and AIS-31 tests are reported and discussed in Section IV. In Section V a comparison against the state-of-the-art TRNGs is carried out. Finally, conclusions are drawn in Section VI.

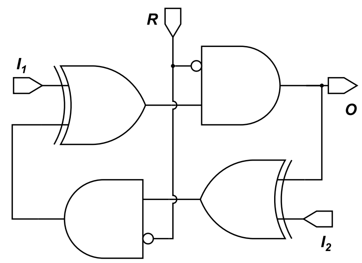

Fig. 1. Latched-Xor (LX) cell.

II. Proposed TRNG Architecture

This Section is organized as follows: In Sec. II-A the LX-cell is introduced, in Sec. II-B the technique used to improve the intrinsic entropy of the LX-cell is outlined and finally in Section II-C transistor level simulations are shown.

A. The LX-Cell as Basic Building Block of the TRNG

The proposed TRNG architecture is based on a novel building block, called the Latched-XOR (LX) cell, recently exploited as key generator in [23] and depicted in Fig. 1. The LX-cell exploits two XOR gates connected through two AND-gates, hence building a fully symmetrical circuit. The LX-cell can be configured to exhibit different behaviours according to the setting of the input signals $I_{1,2}$ and R which is the (active high) reset signal.

The four possible configurations of the LX-cell are shown in Fig. 2:

1) when the R signal is '1' the output of the AND-gates are overwritten and a well defined reset configuration is guaranteed at each net, according to Fig. 2a;
2) when R signal is '0' and $I_1 = I_2 =$'0', both the AND and the XOR gates behave as buffers, as depicted in Fig. 2b;
3) when R is '0' and $I_1 = \overline{I_2}$ the LX-cell behaves as a ring oscillator, as depicted in Fig. 2c, whose oscillation frequency is given by

$$T_{ROi} = \frac{1}{2 \cdot \left(2\, t_{XOR} + 2\, t_{AND}\right)} \tag{1}$$

where t_{XOR} and t_{AND} denote the propagation delay of the XOR and of the AND gate respectively.
4) when R is '0' and both I_1 and I_2 are '1' the LX-cell is equivalent to a SRAM-cell, as depicted in Fig. 2d.

The LX-cell has been employed as the fundamental entropic source in the LX-TRNG. The excitation sequence is depicted in Fig. 3. A detailed explanation of the excitation sequence is presented below:

1) R='1' and $I_{1,2}$= '0', a well defined reset configuration is forced, according to Fig. 2a;
2) R='0' and $I_{1,2}$= '0', the architecture holds nets' values;

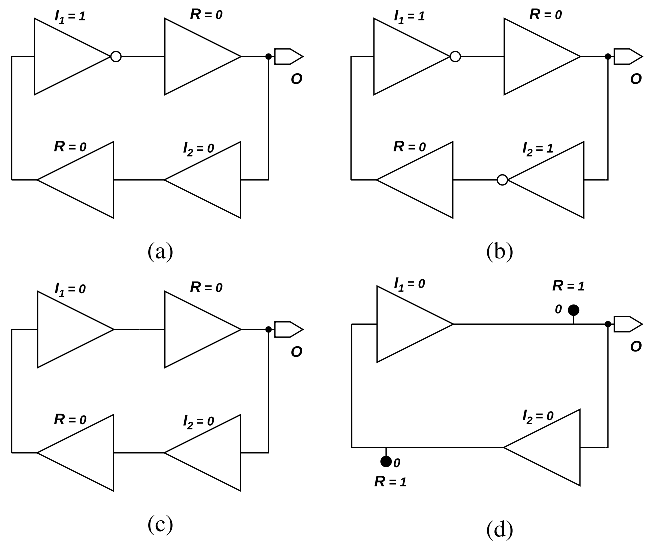

Fig. 2. LX-cell in Reset configuration (a), hold configuration (b), ring-oscillator configuration (c), and SRAM configuration (d).

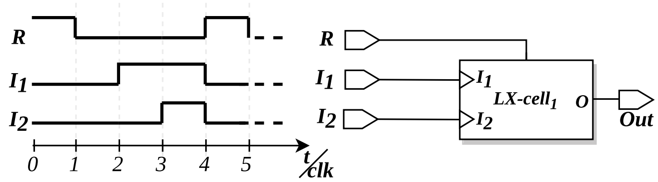

Fig. 3. Excitation sequence of the LX-cell.

3) R=‘0’ and $I_1 = \overline{I_2}$, the TRNG behaves as a ring oscillator;
4) R=‘0’ and $I_1 = I_2$; depending on jitter of each ring-oscillator and on the capability of logical gates to sample a stable data, a stable or metastable output is collected;
5) R=‘1’ and $I_1 = I_2$ =‘0’; the output is collected by a first-in first-out (FIFO) memory, and the architecture can be reinitialized so that a new excitation sequence can be started and a new bit can be generated.

It has to be noted that the LX-cell does not only exploit jitter accumulation but also the noisy behavior of latched XOR gates which in the configuration with R=‘0’ and $I_1 = I_2$ collect a stable or metastable data. In addiction, we want to remark that if compared with the entropic cell of [10] the one adopted in this work is made up only of combinatorial gates (no Flip-Flops are used) and exploits a totally different excitation sequence.

B. Randomness Performance Improvement Technique

The most relevant goal in the design of TRNGs is the optimization of the tradeoff between the entropy that the physical implementation would generate and two other crucial figures of merit: the throughput and the resource usage. In particular, the minimization of hardware resources can be a strict requirement in lightweight applications and could lead to physical sources of entropy that, especially in architectures

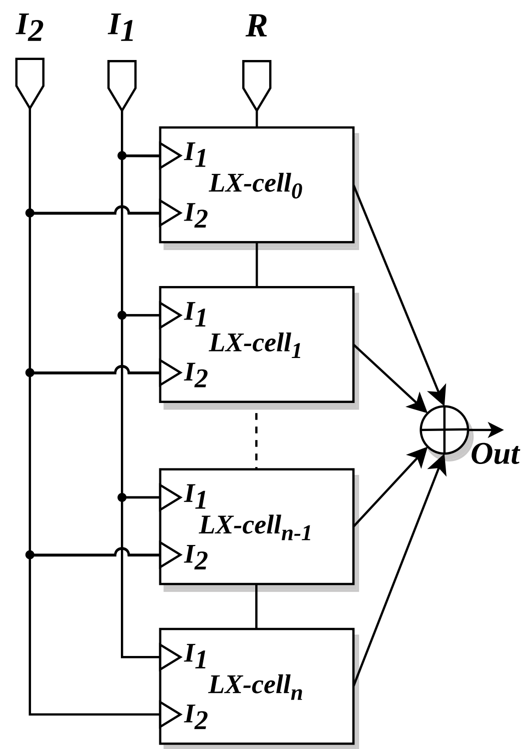

Fig. 4. Scheme of the proposed technique to improve the randomness behaviour.

leveraging on jitter accumulation [5], [9], [10], [16], would require a very long time to attain adequate randomness quality. In some cases, some post-processing techniques should be used to improve randomness performance at the cost of throughput (e.g. Von Neumann Correction.) As an alternative approach, one can exploit N identical cells by XORing the extracted outputs so as to increase the bitsream entropy without reducing the throughput, but at the cost of increased resources usage. Therefore, it is clear that a trade-off between hardware resources and throughput should be satisfied with respect to a given basic cell for entropy generation whose performance strongly relies on the hardware implementation. According to these considerations, we have investigated the LX-cell and its physical implementation and, after preliminary studies and measurements, we have concluded that, as many other lightweight entropy sources in the literature [10], [19], [24], the single LX-cell is not able to generate enough entropy to reach the minimum requirements which NIST or AIS-31 assessment tests demand for. Therefore, with the aim of improving the randomness and entropy performance of the TRNG, we have chosen to collect a bitstream at higher frequency at the cost of hardware resources by exploiting the XOR combining technique as shown in Fig. 4. Referring to Fig. 4, we have studied the improvement of the bitstream entropy leaving unaltered the stimulation sequence depicted in Fig. 3 for each LX-cell.

In this work the goal is to achieve a TRNG that produces good randomness at high throughput. In this regard, the effects of jitter and metastability on the cells when the pulse width is varied have not been investigated since we want to design an high TP TRNG which thus exploits only one unit of pulse width (i.e. one clock cycle) in the excitation sequence (see Fig. 1c). Thus the focus of the work is on the overall approach to combine multiple unbiased cells ($p_{o_i} \neq 0.5$) to generate a bitstream which result biased. Since the quality of the extracted bitstream from each LX-cell depends on the jitter accumulated in a clock cycle (see the excitation sequence in Fig. 3), the extracted sequence contains a number of ones and zeros that follows a Binomial distribution whose parameters are influenced by the accumulated jitter, the metastability and the physical implementation of the cell. Thus, in the following model we want to remark that the metastability and jitter dependence on the pulse width of the excitation signals have not been modeled and the bitstream performance have been considered "at it is" since our focus is on the overall technique and not on the specific entropic cell (i.e. the LX-cell). As a consequence, each bit of the bitstream can be considered as a random variable that follows a Bernoulli distribution in which the '1' event has probability p_0 whereas the '0' event has probability $1-p_0$. If the basic cell exploited as entropic source is unbiased (i.e. $p_0 \neq (1-p_0)$), p_0 is not close to 0.5, and the proposed combining approach can be exploited to obtain a TRNG with p_0 close to 0.5 as required by many cryptographic applications [25], [26], [27].

In the following we assume, without loss of generality, that the physical implementation of the $i-th$ LX-cell would produce, at each experiment, a bit whose event '1' has probability p_{0_i}. Moreover, we suppose that bitstreams extracted at the same time from different LX-cells are uncorrelated. In addiction, we assume that all the N physical implementations of the LX-cell would produce a bitstream whose '1' events are equally probable (i.e. $p_{0_i} = p_0 \; \forall i \in [0, N]$). These approximations and considerations will be better discussed and justified in Section IV with the help of supporting experimental results.

With regards to Fig. 4, it has to be noted that the final XOR operation is carried out with n, 2 inputs, XOR gates, according to Fig. 5c, in which p_0 is the '1' event probability of an LX-cell and p_{o_i} is the '1' probability given by the i-th XOR-gate. With these assumptions the probability p_{out} of event '1' at the output of the architecture can be derived as:

$$p_{out}(p_0, n) = p_0 \sum_{i=0}^{n} (1-2p_0)^i \tag{2}$$

where n is the number of XOR gates employed according to Fig. 5. The proof of the Eq. 2 is reported in Appendix A.

Looking at the expression of p_{out} in Eq. 2, we can recognize a geometric series with parameters n and p_0. Since $p_0 \in (0, 1)$, it follows that $-1 \leq 1-2p_0 \leq 1$ and therefore the p_{out} can be expressed as:

$$p_{out} = p_0 \frac{1-(1-2p_0)^{n+1}}{1-(1-2p_0)} = \frac{1-(1-2p_0)^{n+1}}{2} \tag{3}$$

It has to be noted that, referring to the architecture in Fig. 4, a number n of XOR gates corresponds to $N = n+1$ LX-cells involved. Furthermore, it can be observed, from Eq. 3, that, if n tends to $+\infty$:

$$\lim_{n \to +\infty} p_{out}(p_0, n) = 0.5 \tag{4}$$

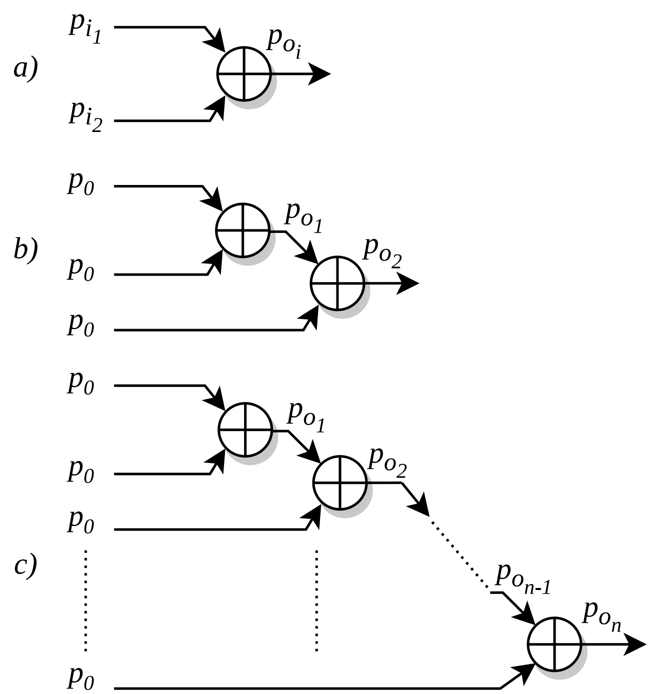

Fig. 5. XOR configuration with two generic inputs (a), XOR configuration with three identical inputs (b) and XOR configuration with N=n+1 identical inputs (c).

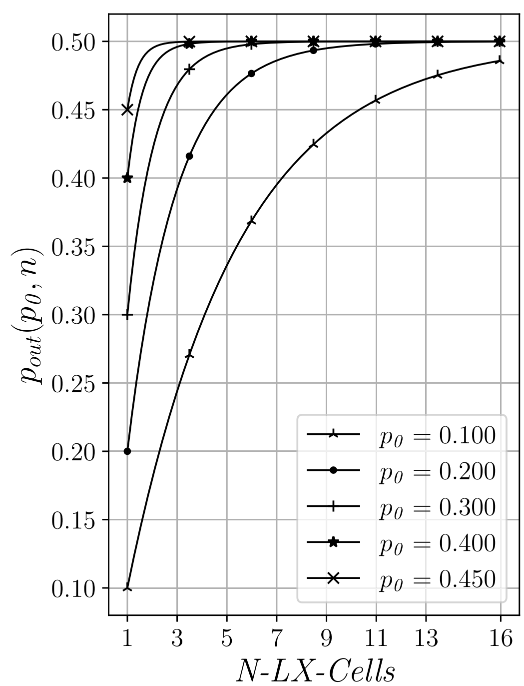

Fig. 6. Trend for given p_0 values.

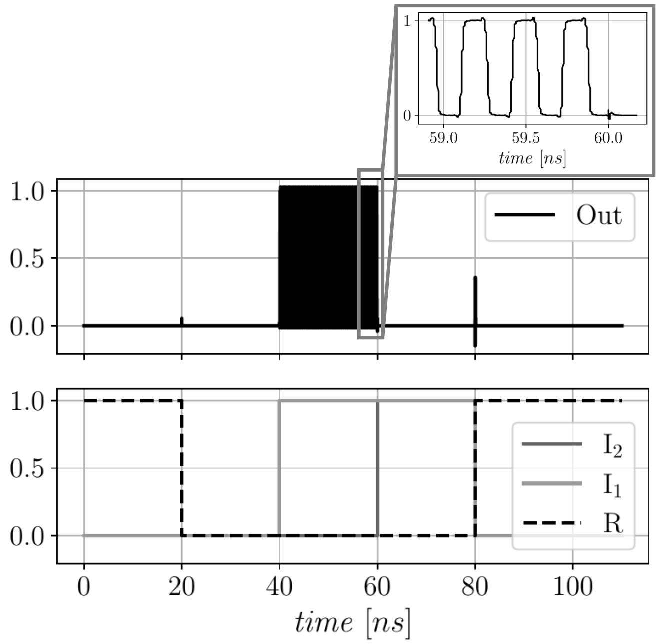

Fig. 7. Cadence simulation of the LX-cell.

Results are in agreement with [28], in which the logical XOR is used as a cryptoanalysis method for the Data Encryption Standard (DES) cipher. From the above analysis it is clear that, given a physical implementation whose probability of event ‘1’ (p_0) is far from the ideal value 0.5, it is possible to produce an output bitstream whose probability of event ‘1’ (p_{out}) is close to 0.5 by using N identical cells whose outputs are combined together through n XOR gates. The minimum number of XOR gates can be selected according to the following equation:

$$n = \frac{log_2(1 - 2p_{out})}{log_2(1 - 2p_0)} - 1 \tag{5}$$

in which the value of p_{out} is directly related to the theoretical $H(p_{out}) = \hat{H}$ which has to be attained. To better clarify this point, we have reported in Fig. 6 the trend of $p_{out}(p_0, n)$ for different p_0 values. As it can be observed, for p_0 values close to the ideal case (i.e. $p_0 = 0.5$) a good entropy can be reached with few XOR gates (meaning few LX-cells). It is also worth to note that, increasing n, a good entropy can be reached even for p_0 values very far from 0.5.

C. LX-Cell Simulation Results

In order to investigate the TRNG architecture based on N LX-cells and n XOR gates, circuit simulations have been carried out through Cadence tools referring and adopting a 55 nm standard-cell CMOS library from ST-Microelectronics. The stimulation sequence which has been used to test the working principle of the architecture, assuming a 50 MHz (20 ns period) clock signal, is reported in Fig. 7.

As it can be observed:

1) At 20 ns the LX-cell is (re-)initialized and the output goes to ‘0’;
2) At 40 ns the LX-cell is set in RO-Configuration because of $I_1 = \overline{I_2}$, and the output starts to oscillate according to the intrinsic delays of used standard cells;
3) At 60 ns the LX-cell latches the value of the last oscillation. It follows that a metastable or stable value which depends on the accumulated jitter would be collected;
4) At 80 ns, a reset is issued and the initial condition is restablished.

It has to be remarked that the aim of this work is different from the focus of [10], in which the jitter accumulation time was exploited to improve entropy performance and whose purpose was to minimize the resource consumption sacrificing TP. With the aim of evaluating the correlation between the output bits of N identical LX-cells, mismatch Monte Carlo simulations have been performed. Simulations pointed out that at each experiment the bits extracted from multiple sites follow a Bernoulli distribution whose probability of event ‘1’ is about $\mu \approx 0.4980$ for $N = 500$, thus confirming that each LX-cell would produce an output that depends on both mismatch variations and jitter accumulation. From the above results, we expect that, if N identical LX-cells are implemented in different FPGA sites, the cross-correlation between the bitstreams extracted from them would be very low and thus the bitstream extracted from each LX-cell would be unique. Furthermore, a well balanced routing of the LX-cells with interconnections as symmetric as possible would increase the TRNG’s performance in terms of randomness, since no-biasing or periodic information can be detected in the generated bitstreams. Indeed, the symmetry of the implementation is an essential feature that has to be achieved in order fully exploit the different entropy sources such as transient noise, Jitter accumulation and cell metastability, and to increase the uncorrelation between multiple LX-cells. As a matter of fact, since the randomness performance relies on mismatch and process variations, a highly symmetrical implementation will guarantee that the bitstream produced by each LX-cell will exhibit the same start-up behaviour and, at the same time, each sample will be uncorrelated with the others, thus as will be better shown later, the randomness performance of the whole TRNG will be increased.

III. LX-TRNG Hardware Implementation

The LX-cell has been integrated on a Xilinx Spartan-6 XC6SLX16 device [29] and the Xilinx ISE 14.7 environment has been chosen as development toolchain. With the aim of improving bitstreams quality in terms of randomness and entropy, an in-depth study on the Xilinx’s FPGA architecture has been carried out so as to maximize branch symmetries of the design. In fact, as outlined in the previous section, high symmetry is important to minimize the dependence of the LX-cells behaviour on the particular location in which they have been implemented.

A. LX-TRNG FPGA Implementation

The Xilinx Spartan-6 architecture is organized in Configurable Logic Blocks (CLBs), and each CLB is composed by two Slices. Each Slice embeds 4 Look-Up-Tables (LUTs)

TABLE I
INTERCONNECTIONS' DELAYS

Instance	t_{p1} $[ps]$	t_{p2} $[ps]$	$\|t_{p1} - t_{p2}\|^*$ $[ps]$	E.D.* [ps]	E.F.† [MHz]
1	210	188	22	1258	397.46
2	332	359	27	1551	322.37
3	250	223	27	1333	375.09
4	381	380	1	1621	308.45

*Estimated Delay through ISE-FPGA-Editor;
†Estimated Frequency with considering ISE-FPGA-Editor delays report and Spartan-6 Data-Sheet [32].

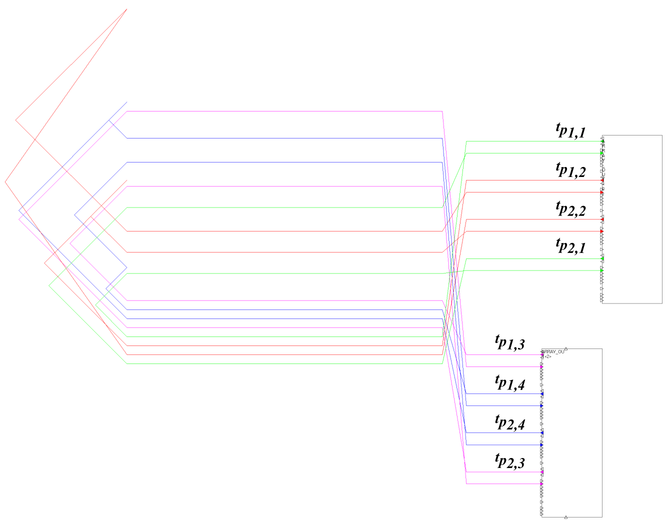

Fig. 8. Spartan-6 implementation of the LX-cell.

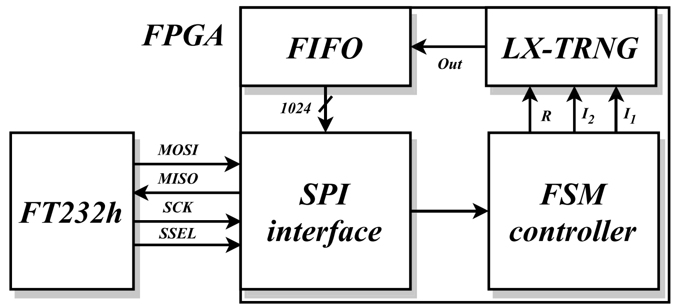

Fig. 9. Block scheme of the TRNG validation testbed.

and 8 Flip-Flops (FFs); inter and intra Slice connections can be managed through the Switch-Matrix. Further details can be found in the official Xilinx User Guide [30]. Since each Slice contains 4 LUTs and each LUT allows to configure logical functions with up to 6 inputs, it follows that each LX-cell can be integrated in just half-slice by configuring two LUTs to implement a XOR and an AND gate with the inversion on one of the inputs (see Fig. 1). However, it has to be noted that, guarantee symmetrical branches to fully exploit mismatch variations between nominally identical LX-cells, is one of the most challenging and at the same time fundamental requirements that should be satisfied in meta-stable latched architectures [23], [31] since the bias of the extracted bitstream would rely on routing symmetries and cells'delays. With this aim, physical placement and routing constraints for the Xilinx Architecture have been exploited and delays of Slices' intra-connections have been evaluated through the Fpga-Editor design tool for each LX-cell. In Tab. I the nominal delays of LUTs' connections that implement the feed of the LX-cells are reported. We have found that the best configuration is obtained with LUT_A connected to LUT_D and LUT_B connected to LUT_C. Pairwised LUTs' connections have been highlighted In Fig. 1 with different colors. After denoting with $t_{p1,i}$ and $t_{p2,i}$ (where i denotes the instance number) the delays of the intra-connections of the Slice from LUT_A to LUT_D and LUT_B to LUT_C of each LX-cells respectively, we have found that the delay differences between the different LUTs intra-connections amount to about 27 ps in the worst case and to about 1 ps in the best case, thus confirming that a very symmetrical configuration has been found. At this purpose, it is worth noting that, with respect to other TRNGs which employ FFs, LUTs, Latches or DCMs, the proposed one is based only on combinatorial gates and thus it is easily portable on almost all FPGA families. Therefore, even if we have exploited Xilinx design tools to balance the delays, we believe that a similar approach can be carried out on almost all FPGA platforms.

The oscillation frequency that each LX-cell exhibits in the RO-configuration (R='0' and $I_1 = \overline{I_2}$) has been evaluated referring to the Spartan-6 Data-Sheet [32] and considering the speedgrade -2. We have found that the delay between the input of the LUTs and the output of the Slice is $\sim$430 ps for each LUT; thereafter the oscillation frequencies produced by the four LX-cells configured in ROs have been evaluated and are reported in Tab. I.

B. Test-Bench

The LX-cell has been tested and characterized through the testbed depicted in Fig. 9. The FPGA uses a system clock of 50MHz, its core is supplied through the Teledyne T3PS43203P, a programmable power supply unit which allows to test the TRNG also under supply voltage variations (nominal core voltage is 1.2 V). The FT232h module has been employed as SPI master and an on-board SPI slave interface has been used to control the Finite State Machine (FSM) which stimulates the LX-cells; generated bits are collected through a 1024-bit first-in first-out (FIFO) memory. The extracted bits are exchanged through the SPI interface and are processed through Python Scripts. The effectiveness of the setup allows

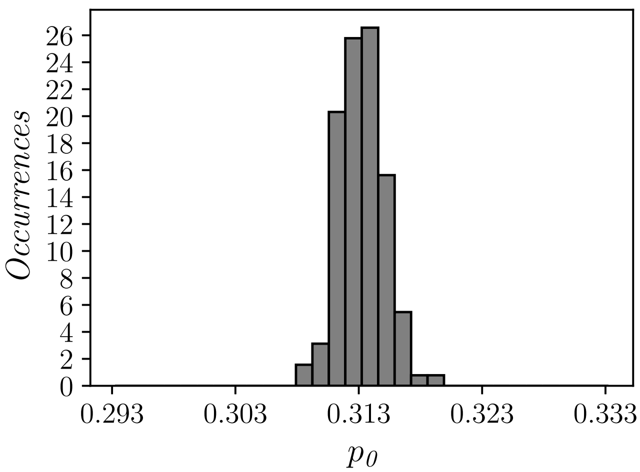

Fig. 10. Trend of the mean value of 10^6 measurements on 128 LX-cells placed in different sites.

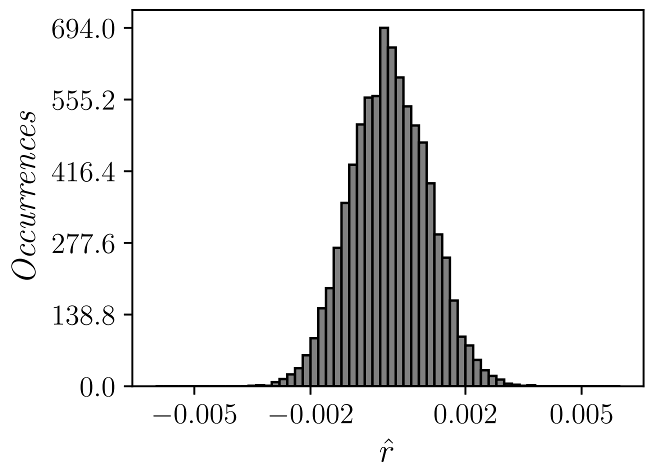

Fig. 11. Histogram of the cross-correlation between the bitstreams extracted from different placed LX-cells.

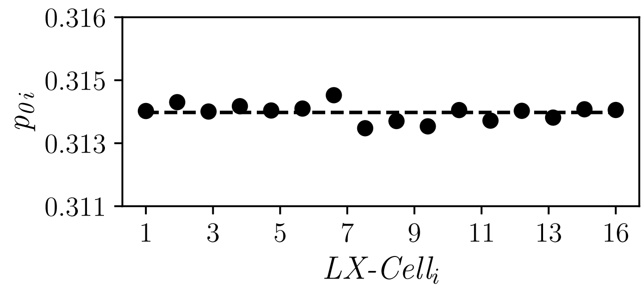

Fig. 12. Bias of the 16 selected LX-cells.

the investigation of the LX-cell and TRNG performance also under supply voltage variations and different stimuli conditions. Furthermore, to evaluate the performance of the TRNG under temperature variations a climatic chamber has been employed as in [10] and [31].

IV. Measurement Results

In this Section the characterization of both the LX-cell and the TRNG are presented. The NIST SP 800–22 suite [25], the NIST SP 800-90B suite [26] and the AIS-31 [27] have been employed to evaluate the bitstream quality and results are reported also under different environmental conditions.

A. LX-Cell Performance Evaluation

In order to evaluate the intrinsic performance of the LX-cell, 128 identical LX-cell macros have been placed in different FPGA's locations in order to investigate the p_{0_i} for each of them. From each LX-cell, 10^6 bits have been extracted and the p_0 value for each of the 128 location has been computed. It has been found that the p_0 values follow a Gaussian distribution with mean value $\mu \approxeq 0.3132$ and $\sigma \approxeq 0.0018$ as depicted in Fig. 10; therefore the standard deviation is about 0.6% of the mean value and the probability of '1' event p_{0_i} can be considered equal to p_0 for each LX-cell, thus confirming the assumptions made in Section II-B. Moreover the cross-correlation of the bitstreams for all the possible permutations have been computed and results are shown in Fig. 11. Measurements of the cross correlation have highlighted a mean value $\mu \approxeq 4.629 \cdot 10^{-5}$ and $\sigma \approxeq 0.99 \cdot 10^{-3}$, thus confirming that bitstreams extracted from different LX-cells are uncorrelated, also in accordance with simulation results in Section II-C and assumptions made in Section II-B. Furthermore, different designs which employ different number of XOR-ed LX-cells have been investigated in order to develop an efficient LX-TRNG and test the working principle and feasibility of the XOR post-processing technique. With this aim, we have evaluated the output of TRNGs with 1 up to 16 LX-cells, and $1024{\cdot}10^5$ bits have been extracted and analysed from each configuration. The p_{0_i} with $i = [1, 16]$ of the $n+1$ LX-cells exploited in this experiment have been reported in Fig. 12. Furthermore, the measured entropies have been compared with those predicted with Eq. 2 and results of these analyses are shown in Fig. 13. Measured results have been found in very good agreement with the theoretical analysis and have confirmed the effectiveness of the proposed approach. According to measurements results, a number of $N = 16$ LX-cells has been selected as the optimal tradeoff between entropy, resources usage and throughput. Thereafter, the performance of the LX-TRNG has been evaluated and results of the analysis have been reported in the following.

It has to be noted that, according to considerations discussed in Section III-A referring to Xilinx FPGAs architecture, it is possible to implement 16 LX-cells in 8 slices. Since each LUT of the FPGA can perform the XOR operation between up to 6 inputs, an additional slice (containing 4 LUTs) is sufficient to implement a 16-inputs XOR function. Therefore, the LX-TRNG reported in Fig. 4 with N=16 can be implemented using only 9 slices.

B. Performance Evaluation of the LX-TRNG in Nominal Condition

The TRNG cell has been evaluated and studied without other post-processing technique (e.g. Von Neumann encoding)

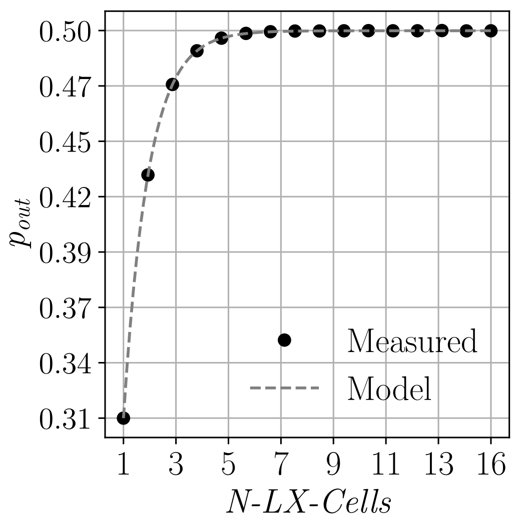

Fig. 13. Predicted and Measured P_{out} of the TRNG architecture for different values of N.

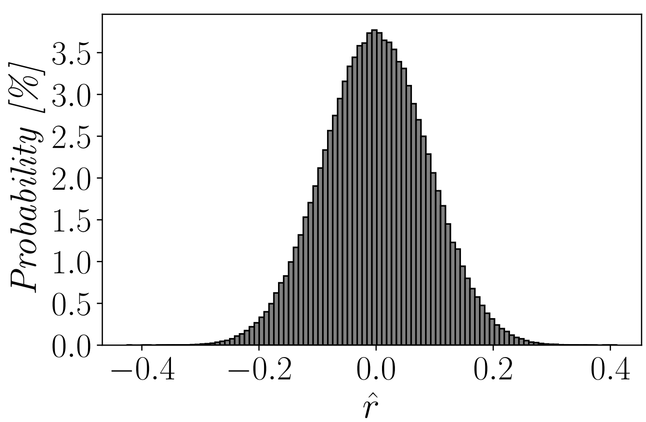

Fig. 14. Correlation between multiple extracted sequences.

TABLE II

NIST SP800-22A TEST RESULTS AT 1.2V

Name of the Test	p-value	pass-rate	result
Frequency Test	0.437	96/100	✓
Block Frequency Test	0.249	97/100	✓
Cumulative Sums$^{\intercal}$	[0.335,0.494]	[96,97]/100	✓
Runs Test	0.575	99/100	✓
Longest Run	0.964	98/100	✓
Rank	0.868	98/100	✓
FFT	0.202	98/100	✓
Non Overlapping	[0.040,0.994]	[96,100]/100	✓
Overlapping	0.740	99/100	✓
Universal	0.658	99/100	✓
Approximate Entropy	0.225	100/100	✓
Random Excursions$^{\intercal}$	[0.074,0.964]	[61/62]/62†	✓
Random Excursions Variant$^{\intercal}$	[0.122,0.834]	[61/62]/62†	✓
Serial	[0.740,0.936]	100/100	✓
Linear Complexity	0.596	99/100	✓

† The minimum pass-rate is 59/62;
$^{\intercal}$ When multiple value are given it has been reported the values' range attained.

with the aim of characterizing only the intrinsic qualities of the LX-TRNG architecture. As a first step we have evaluated the correlation between multiple sequences of 1024 bits extracted by the FIFO in order to assure that the bitstream is not affected by any predictive phenomenon. With this aim, 1000 sequences of 1024 bits have been extracted from the FIFO and according to [10] the correlation (denoted with $\hat{r}$) has been depicted in Fig. 14. We have found that $\hat{r}$ follows a Normal Distribution with mean value $\mu \approx 8.339 \cdot 10^{-5}$ and standard deviation $\sigma \approx 0.0887$. We have also computed the cross entropy over repeated measurements, taken by resetting and reprogramming each time the FPGA. It was found that the mean value of the cross entropy over 100 repeated measurements of 10^5 bytes has a mean value of μ=0.99999971 and a standard deviation σ=7.4634$\cdot 10^{-7}$. Referring to these results, we can state that the sequences extracted in the above experiments are not affected by any correlation. In order to assess the bitstreams quality, 100000 sequences of 1024 bits have been collected. The number of ones and zeros show a Binomial Distribution whereas the mean value of the 1-event occurs with $\mu \approx 0.4998449$, thus the *entropy* of the whole sequence has been evaluated to be around ≈ 0.99999993. However, further analysis should be conducted in order to prove that the TRNG produces complex sequences with enough entropy and random behavior. With this aim, the quality of the extracted sequence has been evaluated through NIST SP 800–22 [25], NIST SP 800-90B [26] and AIS-31 [27] test suites which are among the most used tools in TRNG performance evaluation.

The LX-TRNG passes SP800-22a NIST tests without any post-processing technique and with a throughput that amounts to $TP = 12.5\ Mbit/s$. Furthermore, the generated bitstreams have also been tested with SP800-90B NIST tests and results of the IID test, the χ^2 test, the longest repeated substring test as well as the byte and bit minimum entropy are depicted in Tab. III. Also in this case, all the tests pass successfully. The sequence has been evaluated also with AIS-31 tests and also in this case we have found that the LX-TRNG passes all the tests. Moreover, the result of the T8 test has been exploited to evaluate the entropy of the TRNG and we have found that the estimated Byte-Entropy amounts to ≈ 7.99791; which is a noteworthy result, also if compared with other works, as will be better discussed in the following section.

In order to evaluate the randomness performance also under supply voltage variations, NIST and AIS-31 tests have been performed in a range of $\pm 10\%$ of the nominal supply voltage and results of the analysis are depicted in Tab. IV and Tab. V. In all the considered cases NIST and AIS-31 tests pass and we consider this a noteworthy result. In fact, considering that we are not using any post-processing technique, achieving good performance also in conditions far from the nominal ones is a very challenging task.

The results of the AIS-31 suite T8-test under temperature and voltage variations are reported in Tab IV. Analysing these results, it can be observed how at higher core voltages correspond higher estimated entropy (according to T8 test) and also at higher temperatures the entropy get better. This behavior is probably due to the working principle of LX-cells

TABLE III
NIST SP800-90B RESULTS AT 1.2V

Test			Result C[i][0]	C[i][1]	C[i][2]	Pass
Excursion			21	0	6	✓
NumDirectionalRuns			10	0	6	✓
LenDirectionalRuns			0	6	6	✓
NumIncreasesDecreases			27	0	6	✓
NumRunsMedian			6	0	6	✓
LenRunsMedian			11	3	3	✓
AvgCollision			6	0	6	✓
MaxCollision			5	1	32	✓
Permutation tests	Periodicity	(1)	24	0	6	✓
		(2)	27	0	6	✓
		(8)	6	60	36	✓
		(16)	6	0	43	✓
		(32)	6	0	66	✓
	Covariance	(1)	24	0	6	✓
		(2)	33	0	6	✓
		(8)	11	0	6	✓
		(16)	6	0	7	✓
		(32)	26	0	6	✓
	Compression		15	0	6	✓

Test		Result Score	degrees of freedom	p-value	Pass
χ^2 tests	Independence	65024.365590	65280	0.760080	✓
	Goodness of fit	2234.698875	2295	0.812760	✓

Test		Result
Entropy	H-bitstring	0.999185
	H-original	7.967957
Longest Repeated Substring test		✓

TABLE IV
ESTIMATED BYTE-ENTROPY BY T8-TEST OF AIS-31

Environmental Condition	**1.08V , 25°C**	**1.2V, 25°C**	**1.32V, 25°C**	**1.2V, 50°C**	**1.2V, 75°C**
Estimate Byte-Entropy	7.99714	7.99791	7.99867	7.99812	7.99831

which rely on ROs, whose accumulated jitter get higher with greater core voltage or temperature, thus allowing the jitter to accumulate in a shorter time and to reach better entropy.

V. COMPARISON WITH THE STATE-OF-THE-ART

An overall comparison of FPGA-compatible TRNGs has been carried out and results are in Tab. VI. In terms of throughput, [11] seems comparable to the proposed work but however it has to be noted that it was evaluated with an Operating Frequency (O.F.) of 100 MHz which is two times higher than the one considered in this work and the resource consumption are much higher than the one proposed. The area usage of the proposed TRNG is comparable with those reported in [10] and [34] that however show lower throughput values. Finally, with regard to the FOM introduced in [10] which summarizes TP performance over operating frequency (OF) and hardware consumption (in terms of Slice), defined as:

$$FOM = \frac{TP}{OF \cdot Slices}\left[\frac{bits}{Slice}\right] \tag{6}$$

we observe that the proposed implementation well addresses the tradeoff between hardware consumption and throughput, resulting lower of just [35] and [38]. However, it has to be noted that the bistream quality of [35] has not been tested under process, supply voltage and temperature (PVT) variations. For the sake of fairness, we would like to remark that cryptographic primitives tested only in typical conditions may result in an optimistic resource consumption estimation, since the extracted bitstream may not be random enough to pass NIST and AIS-31 tests under PVT variations. Indeed, PVT variations may affect the bitstream behavior in a deterministic way and thus, to hide or mask this effect, further resources have to be considered. Moreover, it has to be remarked that the FOM reported in Eq. 6 gives a numerical interpretation of the tradeoff between logical resources (in terms of Slices) and TP normalized to the operating frequency. However, in the above FOM the randomness behavior of the TRNG is not taken into account at all. With the aim of comparing different TRNGs', not only from the viewpoint of the tradeoff between logical resources and TP, but considering also the entropy performance, we introduce the figure of merit, FOM_E defined as:

$$FOM_E = \frac{FOM}{8 - H_8} = \frac{TP}{O.F.\ Slice} \cdot \frac{1}{8 - H_8} \tag{7}$$

where H_8 is the byte entropy estimated through the T8 test of the AIS-31 and TP is the throughput. In this regard the proposed FOM_E gives a numerical representation of the tradeoff between TP, resource consumption and also generated entropy, thus allowing a more comprehensive comparison.

TABLE V

NIST SP800-22A TEST RESULTS. VS V_{DD} VARIATIONS

	1.08 V			1.32 V		
Name of the Test	**p-value**	**pass-rate**	**result**	**p-value**	**pass-rate**	**result**
Frequency Test	0.033	96/100	✓	0.494	98/100	✓
Block Frequency Test	0.514	99/100	✓	0.883	100/100	✓
Cumulative Sums$^{\top}$	[0.012,0.091]	96/100	✓	[0.437,0.720]	98/100	✓
Runs Test	0.798	99/100	✓	0.350	98/100	✓
Longest Run	0.456	100/100	✓	0.851	99/100	✓
Rank	0.419	99/100	✓	0.816	99/100	✓
FFT	0.780	100/100	✓	0.319	99/100	✓
Non Overlapping	[0.015,0.983]	[96,100]/100	✓	[0.014,0.991]	[96,100]/100	✓
Overlapping	0.514	100/100	✓	0.237	100/100	✓
Universal	0.637	99/100	✓	0.202	98/100	✓
Approximate Entropy	0.637	98/100	✓	0.401	99/100	✓
Random Excursions $^{\top}$	[0.031,0.976]	[59,60]/60†	✓	[0.029,0.924]	[55,57]/57†	✓
Random Excursions Variant$^{\top}$	[0.122,0.931]	[59,60]/60†	✓	[0.161,0.924]	[55,57]/57‡	✓
Serial $^{\top}$	[0.475,0.514]	[99,98]/100	✓	[0.883,0.972]	99/100	✓
Linear Complexity	0.779	100/100	✓	0.402	100/100	✓

† The minimum pass-rate is 57/60; ‡ The minimum pass-rate is 54/57;
$^{\top}$ When multiple value are given it has been reported the range of the attained values;

TABLE VI

COMPARISON TABLE

Mode Ref	Area$^{\diamond}$				O.F. [MHz]	TP [Mbit/s]	Byte Entropy‡	Bit Entropy*	T range [°C]	V_{DD} range [%]$^{+}$	FOM $\left[\frac{bits}{Slice}\right]$	$FOM_E\left[\frac{bits}{Slice}\right]$	Platform
	LUTs	Flip-Flops	Carry-Chain	Slices									
LX (this work)	32/36$^{\curlywedge}$	0	0	8/9$^{\curlywedge}$	50	12.5	7.9979	0.9997	[25,75]	[-10,+10]	$27.77\cdot10^{-3\curlywedge}$	**13.228**$^{\curlywedge}$	Spartan-6
TROT [33]	32	55	17	33	125	12.5	7.996	0.9995	[-20,70]	[-10,TYP]	$3.03\cdot10^{-3}$	0.757	Zynq-7000
ES [34]	7	6	1	4	100	10	-	-	TYP	TYP	$2.5\cdot10^{-3}$	-	Spartan-6
LRO [10]	4	3	0	**1**	50	0.76	**7.9983**	**0.9998**	[25,75]	[-10,+10]	$15.15\cdot10^{-3}$	8.992	Spartan-6
RO [8]	528	177	0	270	24	6	7.9946	0.9993	[25,75]	TYP	$0.26\cdot10^{-3}$	0.048	Spartan-3A
RO [16]	712	753	0	-	-	3.2	-	-	TYP	TYP	-	-	Spartan-3E
RO [9]	10	5	1	3†	100	1.15	7.9760$^{\top}$	0.9970$^{\top}$	TYP	TYP	$3.83\cdot10^{-3\dagger}$	0.160	Spartan-6
RO [5]	521	131	0	-	-	2.57	7.9920	0.9990	TYP	TYP	-	-	Spartan-6
DNO [35]	15	0	0	4†	100	100	-	-	TYP	TYP	$\mathbf{250\cdot10^{-3}}$	-	Artix-7
Meta-stability [36]	56	19	0	14	400	100	-	-	[0,80]	[-10,+10]	$17.86\cdot10^{-3}$	-	Virtex-6
Meta-stability [37]	-	-	-	797	100	79.18	7.9952	0.9994	TYP	TYP	$0.99\cdot10^{-3}$	0.206	Artix-7
Meta-stability [11]	-	256	-	580	100	12.5	-	-	TYP	TYP	$0.22\cdot10^{-3}$	-	Virtex-4
Meta-stability [12]	128	-	-	-	-	2	-	-	TYP	TYP	-	-	Virtex-5
STR [24]	616	616	0	154†	400	100	7.9760$^{\top}$	0.9970$^{\top}$	TYP	TYP	$1.623\cdot10^{-3}$	0.068	Virtex-5
MSFRO [17]	25	2	0	7	-	**290**	-	-	[0,80]	[-20,+20]	-	-	Virtex-6
Jitter [38]	-	-	-	32	63$^{<}$	190	-	-	[0,80]	[-20,+20]	$94\cdot10^{-3}$	-	Virtex-6

*Bit Entropy computed from the H_8; $^{\curlywedge}$Extra resources considered for the XOR combining function; *Flip-Flops used as latches; $^{\diamond}$Area of the TRNG cell without considering control logic; †Estimated considering the Xilinx Spartan-6 or Artix-7 CLBs [30]; $^{\top}$Estimated ; ‡Estimated with T8 test of the AIS-31; $^{+}$ Tested supply voltage relative variations in percentage; $^{<}$ Estimated from the Operating Period.

In Tab. VI it has been evaluated the FOM_E for all the TRNGs which report the Byte Entropy, the O.F. and the TP and as it can be observed, the generated TP combined with the high entropy attained, results in the highest FOM_E and only the TRNG in [10] exhibits comparable results. However, though [10] results in minimum area consumption, the TP is low due to the fact that the bitstream has to be extracted after having accumulated enough jitter to pass NIST's tests. Since in this work we focused on high throughput performance, we exploited multiple cells that, taken individually, results in minimum resource consumption and when combined together can reach an high throughput behavior with a minimum resource overhead, thus resulting in state of the art FOM_E. Furthermore, it has to be noted that [17] results in lower resource consumption and very high TP if compared with this work. However, authors did not test the architecture with AIS-31 test suite and did not declare clearly an O.F. thus it is impossible to compute the FOM and FOM_E. Though it was not trivial to compare our results with those reported in [17] we tried our best and we introduce a further FOM which neglects the O.F. and substitutes the H_8 with the $H1_{min}$ which denotes the minimum entropy per bit given by the SP 800-90B NIST test. Thus the FOM_N is defined as follows:

$$FOM_N = \frac{TP}{Slice}\frac{1}{(1 - H1_{min})}\left[\frac{Mbit/s}{Slice}\right] \tag{8}$$

The FOM_N computed for the TRNG in [17] results to be 780.331 $\frac{Mbit/s}{Slice}$ since a $H1_{min} = 0.946909$ is reported. The FOM_N computed for the proposed LX-TRNG amounts to 1704.158 $\frac{Mbit/s}{Slice}$, showing how the proposed architecture outperforms also the TRNG in [17]. In addiction, also [34] doesn't report the H_8 from AIS-31 test suite, but report the $H1_{min} = 0.815$ extracted from SP 800-90B NIST test, thus its FOM_N can be computed and results to amount to 13.513, thus is lower than the one achieved by [17] and this work. Also [38] does not report the T8 test of AIS-31 but report the minimum bit entropy as 0.93145, thus its FOM_N is 364.70, outperformed by [17] and this work.

VI. Conclusion

In this paper we have presented a TRNG architecture which combines the effect of latches metastability and ROs jitter to improve entropy and optimize the tradeoff between TP and resources usage. The FPGA implementation occupies only 9 Slices and the TP is 12.5 Mbit/s. The comparison against the state of the art of FPGA-compatible TRNGs has shown that the proposed TRNG outperforms previous works in terms of the tradeoff between TP, bitstreams quality and resources. Furthermore, being implemented by using only combinatorial basic elements, we believe that it can be implemented on almost all FPGA devices available in the market. However, it has to be noted that, the performances of the proposed TRNG architecture are fully optimized if all the LX-cells are identical between each other. At this purpose it is important to carry out an accurate place and route of the LX-cell macro so as to Xor only random information (Jitter) and not a periodic information.

Appendix

A. Output Probability of 1-Event With n-XORed Gates

With referring to Fig. 5a we denote with p_{i_1} and p_{i_2} the two 1-event probability of the $i-th$ XOR-gate. Thus the output 1-event probability can be expressed as:

$$p_{o_i} = p_{i_1}(1 - 2\ p_{i_2}) + p_{i_2} \tag{9}$$

Assuming with p_0 the 1-event probability of a generic LX-Cell, the output probability of the first XOR-gate in Fig. 5b can be derived as:

$$p_{o_1} = p_0(1 - 2p_0) + p_0 \tag{10}$$

With referring to a two levels configuration, the output probability can be expressed as:

$$p_{o_2} = p_0(1 - 2p_0)^2 + p_0(1 - 2p_0) + p_0 \tag{11}$$

It follows that for a three stage the output 1-event probability can be expressed as:

$$p_{o_3} = p_0(1 - 2p_0)^3 + p_0(1 - 2p_0)^2 + p_0(1-2p_0) + p_0 \tag{12}$$

it would be concluded that the overall output probability for $n+1$ XOR-ed cells through n XOR-gates can be written as:

$$p_{o_n} = p_0 \sum_{i=0}^{n} (1 - 2p_0)^i \tag{13}$$

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